



Fig. 6: GUI program output



Fig. 7: Example program with various inputs

are represented by two different amplitudes of sine waves. A high-amplitude sine wave represents the digital 1's and

low-amplitude sine wave represents the digital 0's.

For demodulation of the ASK signal and regeneration of the digital output, the concept shown as a block diagram in Fig. 2 is implemented.

Frequency shift keying. In PSK, as shown in Fig. 3, the 1's and 0's are represented by two frequencies, i.e., 1's for a high-frequency signal and 0's for a low-frequency signal. Because of these two different frequencies, the

scheme provides high-level security and low-noise transmission.

For demodulation, generally PLL

(phase-locked loop) is used to lock the two different frequencies in the desired range and then use them to regenerate the desired binary data. In this program search algorithm has been implemented to replicate the idea of PLL.

Phase shift keying. In PSK (also called binary PSK), the 1's and 0's are represented by a phase shift, i.e., 1's for a sine wave signal and 0's for a 90-degree phase shift of the signal (in other words, a cosine wave). The PSK scheme is shown in Fig. 4.

Software program

An elegant graphic-user-interface (GUI) is presented here for the digital communication design using MATLAB. Also, the readers can learn the use of graphics in MATLAB in a user-friendly way.

Almost each code line has been commented in detail, as much as possible. The '%' symbol separates the code and comments.

An important function to be kept

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